Open Source Gonioreflectometer: a four-axis BRDF instrument and 3D scanner built from printed parts without soldering

Aleksander Norman[a,*]

[a] *California State University, Dominguez Hills, Carson, CA, USA*

Abstract

The Open Source Gonioreflectometer measures the bidirectional reflectance distribution function (BRDF) of flat samples and scans objects in 3D. A light arm on a 305 mm ring and three sample axes (yaw, tilt, spin) set every illumination and viewing direction relative to a fixed camera, following Ansari-Asl et al. (2024). All 58 structural parts are 3D printed in ABS; the 48 bought parts cost US$511.60 from three distributors. No soldering, drilling, sawing or crimping is needed. Calibration uses a mirror for angles, an exposure series for camera response and a pressed-PTFE tile for absolute reflectance. Fitted Lambert + GGX parameters export to game engines. **[TBD: repeatability, accuracy vs. reference data]**.

Keywords: BRDF, gonioreflectometer, material appearance, open hardware, 3D printing, photometric calibration

*Corresponding author: **[TBD: email]**

Specifications table

Hardware name	Open Source Gonioreflectometer
Subject area	Engineering and materials science; computer graphics (material appearance)
Hardware type	Measuring physical properties and in-lab sensors; imaging tools
Closest commercial analog	Laboratory gonioreflectometers and goniospectrophotometers
Open source license	Hardware CERN-OHL-S-2.0; software GPL-3.0-or-later; documentation CC BY-SA 4.0
Cost of hardware	US$511.60 in parts (cap US$630); tools US$469.76
Source file repository	https://github.com/Normansrule/Gonioreflectometer-Open-Source (DOI at release)

1. Hardware in context

Gonioreflectometers measure how a surface reflects light as a function of incident and reflected direction (Nicodemus et al., 1977). Laboratory instruments reach high accuracy at high cost (Höpe et al., 2012, 2006; Saha et al., 2024); image-based (Marschner et al., 2000; Matusik et al., 2003) and robot-arm (Lyngby et al., 2019) setups lower the cost but still need machining or industrial arms. Consumer-hardware capture (Filip et al., 2014) trades angular coverage for speed. Ansari-Asl et al. (2024) describe a multispectral goniometric device with a light arm and a three-axis sample stage. The Open Source Gonioreflectometer reproduces that geometry with printed parts, stepper motors and a USB camera, for builders without a workshop, consistent with open-hardware practice (Bonvoisin et al., 2017; Pearce, 2012).

2. Hardware description

- **Geometry.** M4 (light arm, 20–324°, 24:1 belt on the ring), M3 (yaw, ±180°, 5:1), M2 (tilt, ±90°, 5:1), M1 (spin, direct drive). Camera fixed at 350 mm; light at 300 mm (Fig. 2). Anisotropic materials need M1 (Ward, 1992).

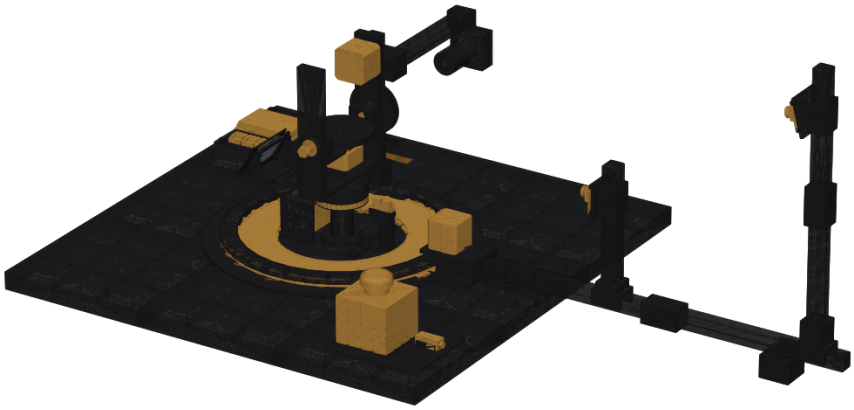


Figure 1: Whole-machine assembly from the repository CAD (`cad/assembly`). Sample axes at centre, light arm on the ring, measuring camera on the right.

- **Light.** Chip-on-board LED, colour rendering index 97, 350 mA constant current, collimating printed tube with a 10 mm aperture; solderless holder clamped by the lamp body.
- **Camera.** 12 MP IMX477 USB module, 16 mm lens, printed f/8 stop; exposure, gain and white balance locked.
- **Control.** grblHAL on an RP2040 board (BTT SKR Pico, USB-C) drives four steppers; a microcontroller reads keypads and sensors; optical homing on every axis; over-travel switches in the emergency-stop loop.
- **One screw size.** Every printed join and module mount is an M2 screw self-tapping into a printed pilot; M3 only into the motors, M8 bolts as axles. The base is a printed deck of nine tiles with a pilot at every mount point.
- **No-tool joins.** Lever connectors, screw terminals and modules (optocoupler, LED driver) instead of loose chips.

3. Design files summary

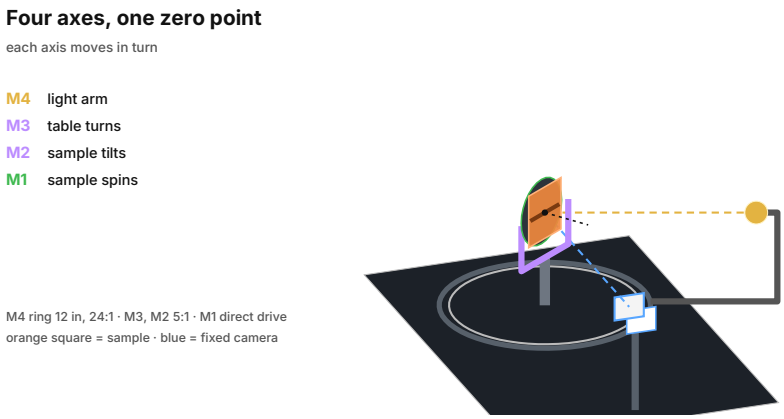


Figure 2: Four axes, one zero point. M4: light arm; M3: table yaw; M2: tilt; M1: spin.

Table 1: Design files summary. Printed parts: `phases/<stage>/3d/`, STEP and STL.

Design file	Type	License	Location
PLAN-01 tolerance coupon	STEP, STL	CERN-OHL-S-2.0	`00-plan`
PWR-18 power plate	STEP, STL	CERN-OHL-S-2.0	`01-power`
CTL-10a button panel	STEP, STL	CERN-OHL-S-2.0	`02-controller`
CTL-10b controller tray	STEP, STL	CERN-OHL-S-2.0	`02-controller`
FRM-01a deck tile SW	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-01b deck tile S	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-01c deck tile SE	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-01d deck tile W	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-01e deck tile C	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-01f deck tile E	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-01g deck tile NW	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-01h deck tile N	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-01i deck tile NE	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-02a outrigger bar 1	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-02b outrigger bar 2	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-02c scan post bar 1	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-02d scan post bar 2	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09c protractor segment	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09d bar saddle	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09e post socket	STEP, STL	CERN-OHL-S-2.0	`03-frame`

Design file	Type	License	Location
FRM-09f end foot	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09h camera pillar	STEP, STL	CERN-OHL-S-2.0	`03-frame`
FRM-09i bar coupler	STEP, STL	CERN-OHL-S-2.0	`03-frame`
LAR-08a rim segment	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
LAR-08b ring motor mount	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
LAR-08c arm segment	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
LAR-08e cross clamp	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
LAR-08f upright bar	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
LAR-08g arm bar	STEP, STL	CERN-OHL-S-2.0	`04-light-arm`
TBL-07a table drum	STEP, STL	CERN-OHL-S-2.0	`05-table`
TBL-07b home flag	STEP, STL	CERN-OHL-S-2.0	`05-table`
TBL-07c table motor bracket	STEP, STL	CERN-OHL-S-2.0	`05-table`
TBL-07d table post	STEP, STL	CERN-OHL-S-2.0	`05-table`
TBL-07f spacers	STEP, STL	CERN-OHL-S-2.0	`05-table`
TBL-07g endstop riser	STEP, STL	CERN-OHL-S-2.0	`05-table`
TBL-07h table flag	STEP, STL	CERN-OHL-S-2.0	`05-table`
CAM-03a aperture stop f8	STEP, STL	CERN-OHL-S-2.0	`06-camera-light`
CAM-03b camera plate	STEP, STL	CERN-OHL-S-2.0	`06-camera-light`
LSR-05a lamp tube	STEP, STL	CERN-OHL-S-2.0	`06-camera-light`
LSR-05b lamp cap	STEP, STL	CERN-OHL-S-2.0	`06-camera-light`
CAL-06a ptfe holder	STEP, STL	CERN-OHL-S-2.0	`07-calibrate`
CAL-06b mirror holder	STEP, STL	CERN-OHL-S-2.0	`07-calibrate`
CAL-06c ball holder	STEP, STL	CERN-OHL-S-2.0	`07-calibrate`
ENC-04a dowel socket	STEP, STL	CERN-OHL-S-2.0	`07-calibrate`
TLT-07a tilt yoke	STEP, STL	CERN-OHL-S-2.0	`08-tilt`
TLT-07b pulley 100T	STEP, STL	CERN-OHL-S-2.0	`08-tilt`
TLT-07c tilt cradle	STEP, STL	CERN-OHL-S-2.0	`08-tilt`
SPN-07a spin platter	STEP, STL	CERN-OHL-S-2.0	`09-spin`
MSK-06a led ring	STEP, STL	CERN-OHL-S-2.0	`10-color`
MSK-06b led mast	STEP, STL	CERN-OHL-S-2.0	`10-color`
SPC-03a spectral tube	STEP, STL	CERN-OHL-S-2.0	`10-color`
SCN-03a riser 20	STEP, STL	CERN-OHL-S-2.0	`11-scan-capture`
SCN-03b riser 40	STEP, STL	CERN-OHL-S-2.0	`11-scan-capture`
SCN-03c riser 60	STEP, STL	CERN-OHL-S-2.0	`11-scan-capture`
SCN-03d scan camera plate 15	STEP, STL	CERN-OHL-S-2.0	`11-scan-capture`
STA-07a pi zero mount	STEP, STL	CERN-OHL-S-2.0	`15-standalone`
STA-10a screen stand	STEP, STL	CERN-OHL-S-2.0	`15-standalone`

Design file	Type	License	Location
STA-10b screen bezel	STEP, STL	CERN-OHL-S-2.0	`15-standalone`
Bought-part stand-ins	STEP, STL	CERN-OHL-S-2.0	`cad/bought`
Whole-machine assembly	STEP	CERN-OHL-S-2.0	`cad/assembly`
CAD source (CadQuery)	Python	GPL-3.0-or-later	`tools/cad`

4. Bill of materials summary

Parts from Digi-Key, Amazon and Adafruit. Tools (US$469.76, including the printer) are listed separately in the repository. Printed parts: ≈2666 g ABS.

Table 2: Bill of materials summary (core machine; shared packs counted once; prices in USD, checked on the store pages).

Designator	Component	No.	Unit	Total	Source
FIL-01	ABS filament 1.75 mm, black, 1 kg (4 spools)	4	15.99	63.96	Amazon
PWR-30	24 V 3 A wall adapter with screw-terminal DC jack	1	17.85	17.85	Amazon
PWR-28	Emergency stop in its own box, latching, 1 NO + 1 NC, screw terminals	1	8.99	8.99	Amazon
PWR-10	Optocoupler module PC817, 24 V input, screw terminals (2-pack)	1	6.99	6.99	Amazon
PWR-15	18 AWG silicone wire red/black, 1.5 m (5 ft) each colour	1	7.98	7.98	Amazon
PWR-20	Lever wire connectors, WAGO 221 assortment (12)	2	12.45	24.90	Amazon
CTL-01	BIGTREETECH SKR Pico V1.0 (RP2040, 4 × TMC2209 UART, USB-C)	1	35.99	35.99	Amazon
CTL-03	USB-A to USB-C data cables, 3 ft (2-pack)	1	8.99	8.99	Amazon
CTL-05	Membrane keypad 1 x 4	1	7.89	7.89	Amazon
CTL-06	JST-XH pre-crimped lead kit	1	10.99	10.99	Amazon
CTL-07	Jumper wires male-female 20 cm (120)	1	6.98	6.98	Amazon

Designator	Component	No.	Unit	Total	Source
FRM-06	M2 socket screw kit, 4–25 mm, with nuts and washers (900 pcs)	1	7.58	7.58	Amazon
FRM-07	Socket screw kit M2/M3/M4 with nuts (820 pcs): M3 for the motors	1	9.88	9.88	Amazon
FRM-08	608-2RS bearings (10)	1	4.99	4.99	Amazon
LAR-01	12 in lazy Susan bearing	1	13.99	13.99	Amazon
LAR-02	NEMA 17 stepper 42 Ncm 1.5 A, 38 mm (5-pack: M4, M3, M2 + 2 spare)	1	32.99	32.99	Amazon
LAR-03	GT2 6 mm open belt (5 m)	1	7.59	7.59	Amazon
LAR-04	GT2 20T pulley 5 mm bore (5-pack)	1	3.99	3.99	Amazon
LAR-05	Optical endstop module, slot type (6-pack)	1	8.99	8.99	Amazon
LAR-06	Roller-lever limit switch, sealed, 3 wires attached (2-pack)	1	9.88	9.88	Amazon
TBL-02	GT2 6 mm closed belt 232 mm	1	8.19	8.19	Amazon
TBL-05	M8 x 75 hex bolt, partial thread + M8 nut (fixed table axle)	1	15.37	15.37	Amazon
CAM-01	USB camera 12 MP IMX477, M12 mount (UVC)	1	59.99	59.99	Amazon
CAM-02	M12 lens 16 mm, $\geq 1/2$ in format, low distortion	1	30.11	30.11	Amazon
LSR-06	Solderless COB holder 13.5 mm (TE LUMAWISE Z35)	1	3.30	3.30	Digi-Key
LSR-01	COB LED CRI 90, 9 V, 13.5 mm (Citizen CLU701)	1	4.60	4.60	Digi-Key
LSR-02	Constant-current driver 350 mA, wire leads (MEAN WELL NLDD-350HW)	1	7.20	7.20	Digi-Key
LSR-03	Heatsink 40 x 40 x 11.6 mm, pin fin (Wakefield 910)	1	10.67	10.67	Digi-Key
ENC-02	Black felt sheet	1	6.99	6.99	Amazon
CAL-01	PTFE sheets 100 x 100 x 3 mm (2): white tile = 3 stacked 50 mm squares	1	16.99	16.99	Amazon
CAL-02	Glass mirror tile ~50 mm	1	11.49	11.49	Amazon
CAL-04	ChArUco print + picture-frame glass	1	4.31	4.31	Amazon
TLT-02	GT2 6 mm closed belt 280 mm	1	7.49	7.49	Amazon

Designator	Component	No.	Unit	Total	Source
TLT-05	M8 x 35 hex bolts, half thread (10-pack; 2 used)	1	9.19	9.19	Amazon
SPN-01	NEMA 17 pancake stepper 16 Ncm 1 A (direct drive)	1	11.99	11.99	Amazon
SCN-01	Mounting putty 2 oz	1	2.33	2.33	Amazon
			Total	**511.60**	

5. Build instructions

The repository splits the build into stages (power, controller, frame, each axis, camera and light, calibration). Each step lists time, parts, actions, a measured pass criterion and what to check on failure (`phases/*/README.md`).

1. Print the tolerance coupon; set clearance and self-tap hole size.
2. Wire the power (wall adapter, e-stop, limit switch); check polarity before the first power-on.
3. Mount the controller; move one motor from the laptop.
4. Print and bolt the deck; screw every mount into its pilot; assemble the axes; home each axis.
5. Mount camera and lamp; run the mirror test; calibrate (Section 7).

6. Operation instructions

1. `opengonio check-pose M4=60 M3=30`: angle and collision check.
2. `opengonio capture inplane -step 10 -out data/s`: in-plane sweep.
3. `opengonio curve data/s`: calibrated BRDF curve and Lambert + GGX fit (Trowbridge & Reitz, 1975; Walter et al., 2007).

Safety: the lamp is viewed only through the sample; the emergency stop cuts motor power.

7. Validation and characterization

Calibration.. Angle offset from the mirror peak ($M_3 = M_4/2$ in-plane); dark level from a lamp-off frame; camera response from an exposure series (Debevec & Malik, 1997); absolute scale from pressed PTFE (Weidner & Hsia, 1981):

$$f_r = \frac{C_{\text{lin}}}{C_{w,\text{lin}}} \frac{\rho_w}{\pi} \frac{\cos 45^\circ}{\cos \theta_i}, \qquad C_{\text{lin}} = (C - C_d)^\gamma. \tag{1}$$

Results.. **[TBD: angular repeatability per axis (mirror, 10 repeats); lamp drift over a run; PTFE reflectance vs. reference; matte and glossy samples vs. MERL (Matusik et al., 2003) or RGL ("RGL-EPFL material database," n.d.) data; scan accuracy]**.

Ethics statements

No human or animal subjects.

CRediT author statement

Aleksander Norman: Conceptualization, Methodology, Software, Hardware, Writing – original draft.

Declaration of competing interest

None.

Acknowledgements

[TBD: acknowledgements]

References

Ansari-Asl, M., Barbieri, M., Obein, G., & Hardeberg, J. Y. (2024). Optical and electromechanical design and implementation of an advanced multispectral device to capture material appearance. *Journal of Imaging*, *10*(3), 55. https://doi.org/10.3390/jimaging10030055

Bonvoisin, J., Mies, R., Boujut, J.-F., & Stark, R. (2017). What is the 'source' of open source hardware? *Journal of Open Hardware*, *1*(1). https://doi.org/10.5334/joh.7

Debevec, P., & Malik, J. (1997). Recovering high dynamic range radiance maps from photographs. *Proc. SIGGRAPH '97*. https://pauldebevec.com/Research/HDR/debevec-siggraph97.pdf

Filip, J., Vávra, R., & Krupička, M. (2014). Rapid material appearance acquisition using consumer hardware. *Sensors*, *14*(10), 19785–19805. https://doi.org/10.3390/s141019785

Höpe, A., et al. (2012). ARGon3: '3D appearance robot-based gonioreflectometer' at PTB. *Review of Scientific Instruments*, *83*(4), 045102. https://pubs.aip.org/aip/rsi/article/83/4/045102/355039/ARGon3-3D-appearance-robot-based

Höpe, A., Hünerhoff, D., & Hauer, K.-O. (2006). Robot-based gonioreflectometer. In *Industrial robotics: Programming, simulation and applications* (pp. 624–632). (ed. L. K. Huat). https://cdn.intechopen.com/pdfs/284/InTech-Robot_based_gonioreflectometer.pdf

Lyngby, R. A., Matthiassen, J. B., Frisvad, J. R., Dahl, A. B., & Aanæs, H. (2019). Using a robotic arm for measuring BRDFs. *SCIA 2019*. https://doi.org/10.1007/978-3-030-20205-7_16

Marschner, S. R., Westin, S. H., Lafortune, E. P. F., & Torrance, K. E. (2000). Image-based bidirectional reflectance distribution function measurement. *Applied Optics*, *39*(16), 2592–2600. https://doi.org/10.1364/AO.39.002592

Matusik, W., Pfister, H., Brand, M., & McMillan, L. (2003). Efficient isotropic BRDF measurement. *Proc. 14th Eurographics Workshop on Rendering*. https://www.pbr-book.org/4ed/Reflection_Models/Further_Reading

Nicodemus, F. E., Richmond, J. C., Hsia, J. J., Ginsberg, I. W., & Limperis, T. (1977). Geometrical considerations and nomenclature for reflectance. *NBS Monograph*, *160*. https://doi.org/10.6028/NBS.MONO.160

Pearce, J. M. (2012). Building research equipment with free, open-source hardware. *Science*, *337*(6100), 1303–1304. https://doi.org/10.1126/science.1228183

RGL-EPFL material database. (n.d.). https://rgl.epfl.ch/pages/lab/material-database

Saha, D., Gevaux, L., Frisvad, J. R., & Obein, G. (2024). Bidirectional reflectance measurements over a micrometric surface area using a goniospectrophotometer. *Review of Scientific Instruments*, *95*(5), 055114. https://doi.org/10.1063/5.0193490

Trowbridge, T. S., & Reitz, K. P. (1975). Average irregularity representation of a rough surface for ray reflection. *Journal of the Optical Society of America*, *65*(5), 531–536. https://doi.org/10.1364/JOSA.65.000531

Walter, B., Marschner, S., Li, H., & Torrance, K. (2007). Microfacet models for refraction through rough surfaces. *Eurographics Symposium on Rendering*.

Ward, G. J. (1992). Measuring and modeling anisotropic reflection. *ACM SIGGRAPH Computer Graphics*, *26*(2), 265–272. https://doi.org/10.1145/142920.134078

Weidner, V. R., & Hsia, J. J. (1981). Reflection properties of pressed polytetrafluoroethylene powder. *Journal of the Optical Society of America*, *71*(7), 856–861. https://doi.org/10.1364/JOSA.71.000856